

Design 3: 2x2 repeated measures design with crossed random effects for subjects and items

Next, we performed SBC for Bayes factors in a 2×2 Latin square design, with crossed random effects for subjects and items. We simulated data from 16 subjects and from 8 trials, leading to a total of $16 \times 8 = 128$ data points. We simulated data from the following model (H1), where n indicates the row of the data frame, $subj[n]$ yields the subject number in row n , and $item[n]$ yields the item number in row n .

$$y_n \sim Normal(\mu_n, \sigma_{Residual}) \quad (28)$$

$$\mu_n = \beta_{(Intercept)} + u_{subj[n],1} + w_{item[n],1} + \quad (29)$$

$$(\beta_{meA} + u_{subj[n],2} + w_{item[n],2}) \times meA_n + \quad (30)$$

$$(\beta_{meB} + u_{subj[n],3} + w_{item[n],3}) \times meB_n + \quad (31)$$

$$(\beta_{int} + u_{subj[n],4} + w_{item[n],4}) \times int_n \quad (32)$$

The random effects u and w are each multivariate normally distributed:

$$\begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \Sigma_u \right), \quad (33)$$

$$\begin{pmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \Sigma_w \right), \quad (34)$$

where Σ_u and Σ_w are the variance-covariance matrices, each with the standard deviation parameters $\sigma_{Random\ slopes}$ and the correlation parameters $\rho_{Random\ slopes}$. The contrast coding for the two main effects and the interaction (meA_n , meB_n , and int_n) was set to sum-coding ($-1/+1$). The prior distributions (from which simulating parameter values were sampled) were the following:

$$\beta_{(Intercept)} \sim Normal(0.7, 0.1) \quad (35)$$

$$\beta_{Fixed\ effects} \sim Normal(0, 0.1) \quad (36)$$

$$\sigma_{Random\ slopes} \sim Normal_+(0, 0.1) \quad (37)$$

$$\sigma_{Residual} \sim Normal_+(0, 0.1) \quad (38)$$

$$\rho_{Random\ slopes} \sim LKJ(2) \quad (39)$$

where $Normal_+$ denotes a normal distribution truncated at zero.

We performed three SBC simulations, one for main effect A, one for main effect B, and one for the interaction $A \times B$. Each simulation used the same H1 model (shown above). In each simulation, the null model (H0) was created by setting one fixed effects parameter to zero. When performing SBC for main effect A, $\beta_{meA} = 0$; when performing SBC for main effect B, $\beta_{meB} = 0$; when performing SBC for the interaction, $\beta_{int} = 0$. In each SBC simulation, we defined prior model probabilities as $p(H0) = 0.8$ and $p(H1) = 0.2$. In each SBC simulation, we fitted the same models to the data that were used to generate the data (i.e., H1 and the corresponding H0). In the model fitting, we used 10,000 iterations, with 2,000 iterations of warm-up. For bridge sampling, we used the “warp3” method. We performed 200 runs of SBC. In additional simulations, we repeated the SBC analyses with 20,000 iterations, again with 2,000 warm-up iterations. Moreover, we performed SBC analysis with 1,000 SBC simulations (10,000 MCMC samples).

Selective reporting

As is widely noted, Bayesian posteriors do not change under data-dependent selective reporting, i.e. for any binary random variable A , such that A is conditionally independent of θ given y , we have $p(\theta|y) = p(\theta|y, A = 1)$. This implies that the expected posterior does not change when we filter datasets and therefore all consistency guarantees, including SBC are valid also for just a subset of the simulations as long as the exclusion criterion is solely a function of the simulated data. Notably this includes any diagnostics from the Bayes factor computation as those only depend on data and it is therefore valid to perform SBC separately for simulations that satisfy a diagnostic check and simulations that fail a diagnostic check.

Results

Design 1: 2x2 design with random effects for subjects

First, we studied a repeated measures 2×2 design, which was inspired by the 2-step decision-making task (Daw et al., 2011; Schad et al., 2014), by running marginal SBC. We investigated model convergence. Across all model fits (200 simulations $\times$ 3 effects $\times$ 2 models (H_0/H_1) = 1200 model fits), we observed 0 divergent transitions. R-hat exceeded the threshold of 1.01 for at least one population-level parameter in 0 cases, suggesting good model convergence. However, during the bridge sampling, there was the warning message “logml could not be estimated within maxiter, rerunning with adjusted starting value. Estimate might be more variable than usual” in 20.33% of simulations, suggesting problems with the bridge sampling. Due to these problems, we looked at the results separately for simulations with versus without warning messages from the bridge sampler.

The results for marginal SBC (see Fig. 1A-B) showed for simulations without warning messages that the posterior model probabilities were equal to the prior model probability of 0.5 for both main effects and for the interaction. Indeed, the null hypothesis that posterior model probabilities were equal to 0.5 was supported by Bayesian t-tests (see Fig. 1B)